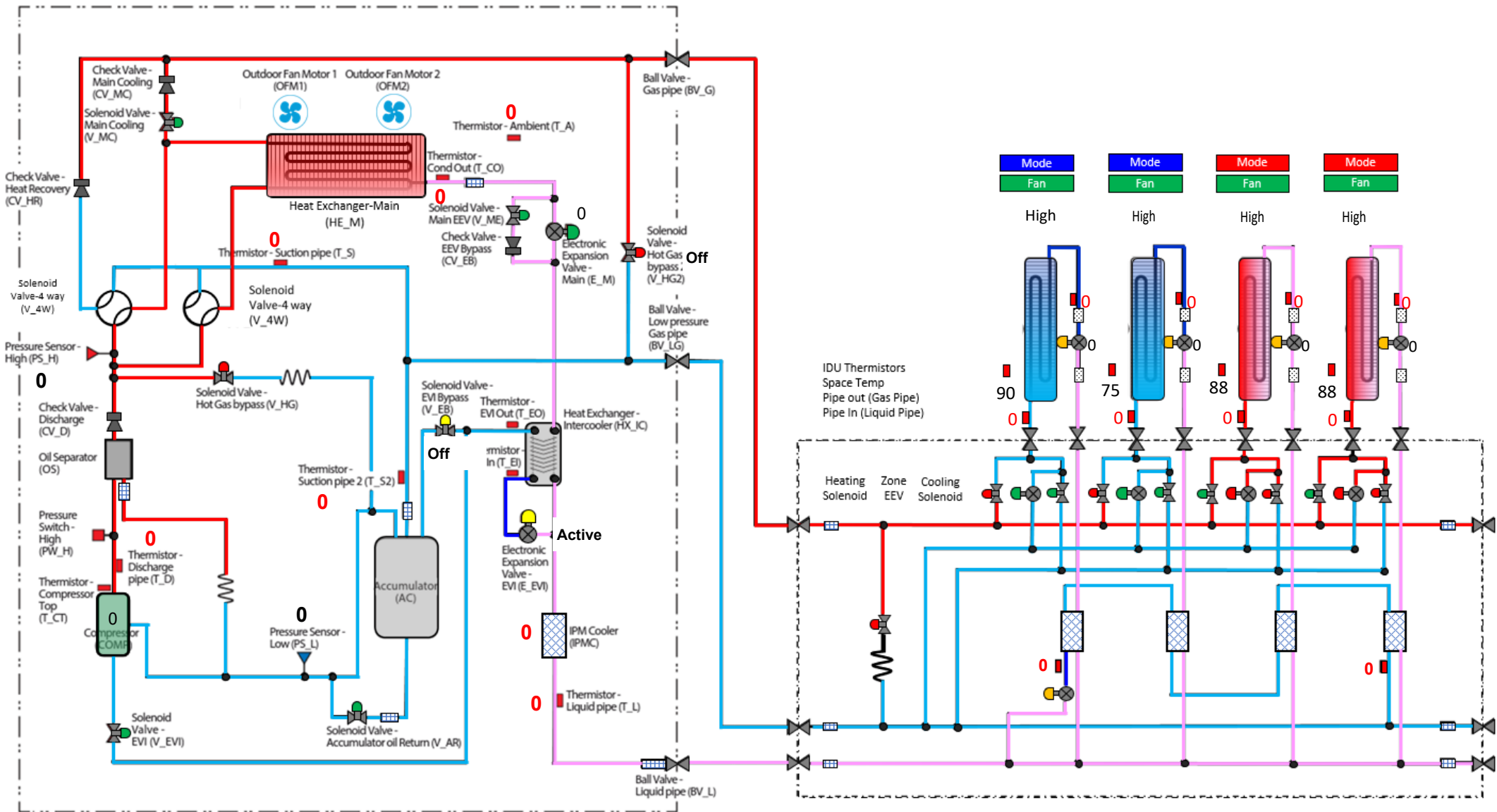


DVM S Heat Pump Operation Evaluation
Cooling Main Mode



Comments:

Outdoor Unit			
Component	Target	Reading	
Ambient Temp.		0	
Compressor Hz.		0	Ok
High pressure	398 psig	0	Ok
H.P Sat. Temp	116 ⁰		
Discharge Temp.	150 ⁰ -212 ⁰	0	Ok
IPM Temp,	Below 194 ⁰	0	Ok
Low pressure	114 psig	0	Ok
L.P. Sat. Temp	38 ⁰		
Suction 1 Temp.		0	Ok
Suction 2 Temp		0	Ok
S.H.	0 ⁰ or 12.6 ⁰	0	Ok
Min EEV Position		0	Ok
Cond. Out Temp.	OAT + (5 ⁰ – 36 ⁰)	0	Ok
Liquid Line Temp.		0	Ok
Sub-Cooling @ T_L	9 ⁰ - 36 ⁰	0	Ok
Indoor Units			
Component	Target	Reading	
IDU 1 EEV	250 – 1400 Steps	0	Ok
IDU 1 SH	3.6 ⁰ (Typical 0 ⁰ -7 ⁰)	0	Ok
IDU 1 T PI		0	Ok
IDU 1 T PO		0	Ok
IDU 2 EEV	250 – 1400 Steps	0	Ok
IDU 2 SH	3.6 ⁰ (Typical 0 ⁰ -7 ⁰)	0	Ok
IDU 2 T PI		0	Ok
IDU 2 T PO		0	Ok
IDU 3 EEV	500 – 1500 Steps	0	Ok
IDU 3 SC	9 ⁰	0	Ok
IDU 3 T PI		0	Ok
IDU 3 T PO		0	Ok
IDU 4 EEV	500 – 1500 Steps	0	Ok
IDU 4 SC	9 ⁰	0	Ok
IDU 4 T PI		0	Ok
IDU 4 T PO		0	Ok

Directions:
Enter the data in the spaces provided. The reading will populate in the refrigerant diagram. Space temperature is entered at the IDU. Use the drop down boxes to identify the state of the component. Use the operating data and the target operation data on the next page to troubleshoot the system.

Air Source Outdoor Unit				
Component	Abbreviation		Cooling Main Mode	Comments
Compressor	Comp	Operation	Target low pressure control (default: 113.8psig)	Primary function, Maintain Average Evaporator Temperature
		Range	14~140Hz	
Outdoor Fan Motor	OFM	Operation	Target high pressure control (default: 398 psig)	Fans speed control is used to maintain head pressure
		Range	0-39 step (0-1300rpm)	Typical Head Pressure Range 355 psig – 469 psig
Main EEV	E_M	Operation	Cooling & Heating capacity balance control	
		Range	300 - 2000 Steps	
EVI EEV	E_EVI	Operation	*Subcooling control (default: SC=36°F) / Discharge Temp. Protection	
		Range	0-480 step	
4 Way Valve	V_4W	Operation	Off	
EVI By-Pass Solenoid Valve	V_EB	Operation	On (Open): *T_a > 100.4°F or all compressors in unit are at 50Hz↓	Activates Sub-Cooling Control
			Off (Close): T_a < 96.8°F and all compressors in unit are at 70Hz↑	Discharge Temperature Control, Control by unit
EVI Solenoid Valve	V_EVI	Operation	On (Open)	
Hot Gas By-Pass	V_HG	Operation	Off (Closed)	
Accumulator Oil Return Solenoid Valve	V_AR	Operation	On (Open when compressor is running)	Off during start up if DSH is < 18°F. Off for 2 min. if DSH is < 18°F On when DCSH > 27°F
Hot Gas By-Pass 2 (HR Only)	V_HG2	Operation	Off Closed	Opens after 20 minutes of continuous run time in cooling mode
Main EEV Solenoid Valve (HR Only)	V_ME	Operation	Off (Closed)	Helps force hot gas to the MCU & IDU's
Main Cooling Solenoid	VMC	Operation	On (Open)	
Condenser Out Thermistor	T_CO	Value	Should be 5-36 drgrees above outside air temperature	Used to calculate condenser sub-cooling
Mode Control Unit (MCU)				
Component	Abbreviation		Cooling Mode	Comments
Port Heating Solenoid	Heat	Operation	On (Open) Call for Heating Off (Closed) Call for Cooling	
Port Cooling Solenoid	Cool	Operation	On (Open) Call for Cooling Off (Closed) Call for Heat	
Port EEV	EEV	Operation	Open 480 Steps for Cooling and 0 Steps heating Operation	Open 480 step when a zone calls for cooling
		Range	0 - 480 Steps	Closed 0 Steps when a zone is in heating opens on mode change
Sub-Cooling EEV	SC_EEV	Operation	Active in Cooling Main	Provide subcooling for cooling zones
		Range	Modulating	Maintains 5.4 degrees of superheat between pipe in and out temps.
Sub-Cooler In Temperature		Operation	Should be colder then Pipe Out	
Sub-Cooler Out Temperature		Operation	Should be hotter then pipe in.	
Indoor Unit IDU Cooling Mode				
Component	Abbreviation		Cooling Mode	Comments
Room Temperature	Room Temp.	Operation	65~86°F	
Liquid Pipe Thermistor	EVA In 1	Operation	Evaporator Temperature	Used to calculate Evaporator Superheat
Gas Pipe Thermistor	EVA Out 1	Operation	Measures Suction Pipe Temperature	Used to calculate Evaporator Superheat
EEV	EEV 1	Operation	Modulates to maintain Superheat of 3.6 degrees F	Superheat = Pipe out Temp - Pipe in Temp (Typical range 0-7 degrees)
		Range	0 - 2000 Steps (0 Steps when zone is satisfied)	Typical range between 250 and 1500 Steps
IDU Fan	Fan Speed	Range	Low , Medium, High, Auto	
		Operation	Low Speed: Ts - Ti ≤ 1.8°F	TS - Set Temperature
			Medium Speed: 1.8°F < Ts - Ti ≤ 5.4°F	Ti Indoor Temperature
			High Speed 5.4°F < Ts - Ti	
Indoor Unit IDU Heating Mode				
Component	Abbreviation		Heating Mode	Comments
Room Temperature	Room Temp.	Operation	67-80 Degrees	
Liquid Pipe Thermistor	EVA In 1	Operation	Liquid Pipe Temperature for sub-cooling Control	Used to calculate Evaporator Sub-Cooling
Gas Pipe Thermistor	EVA Out 1	Operation	Hot Gas Pipe Temperature should be close to discharge temp.	Used to evaluate the temperature of the hot gas coming into the coil
EEV	EEV 1	Operation	Modulates to maintain 9 degrees of sub-cooling	Sub-Cooling = High Side Saturation Temperature - Pipe in Temp.
		Range	0 - 2000 Steps (0 Steps when zone is satisfied)	Typical range between 500 and 1500 Steps
IDU Fan	Fan Speed	Range	Low , Medium, High, Auto	
		Operation	Low Speed: Ts - Ti ≤ 1.8°F	TS - Set Temperature
			Medium speed: 1.8°F < Ts - Ti ≤ 5.4°F	Ti Indoor Temperature
			High Speed 5.4°F < Ts - Ti	